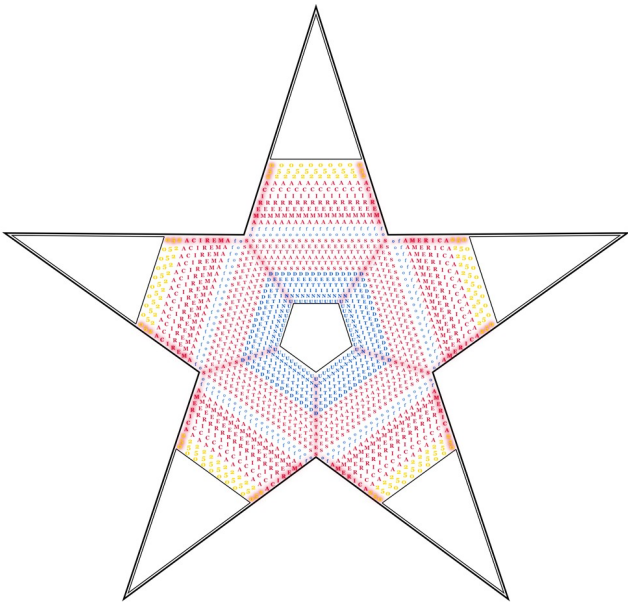


THE GREAT AMERICAN STAR

STARTING POINTS - EDUCATOR GUIDE

Meet The Great American Star

The Great American Star is a mathematical-art project created for America 250. Its structure combines a five-pointed star with a modified Pascal's Triangle, creating a large system of branching and recombining paths.



50	50	359,277,250	1
starting points	ending points	possible paths	American star

Built from a modified Pascal's Triangle

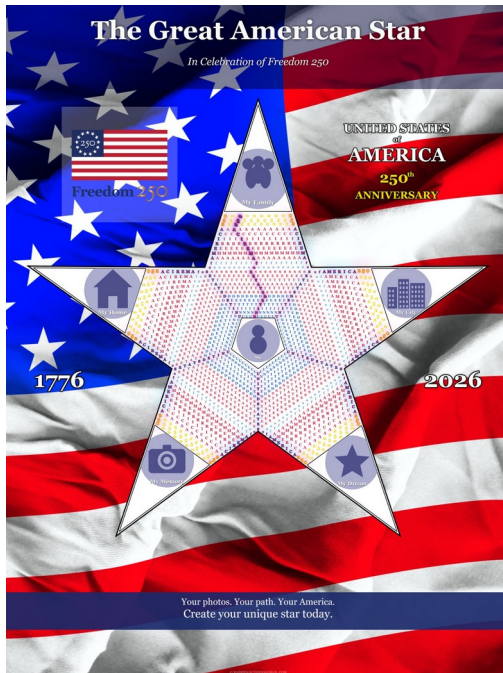
A modified Pascal's Triangle diagram. It shows a triangular arrangement of numbers. The numbers are arranged in rows, with each row starting with a 1. The numbers increase as they move down the rows. Some numbers are highlighted in yellow and red, indicating specific paths or values. The overall shape is a triangle, but it's modified to fit the structure of the Great American Star.

At its mathematical core, the star uses a modified form of Pascal's Triangle - a pattern familiar in many mathematics classrooms.

As paths move through the structure, choices branch and recombine in a Pascal-style pattern. Repeating that structure through the five sections produces 359,277,250 possible paths across the complete star.

1. Mathematics, art, and personal connection

The same mathematical structure can become a personal artwork. A student chooses one path from the hundreds of millions of possibilities and can personalize the six open image spaces in the finished star.



A simple classroom introduction

"This star has 50 starting points and 50 ending points, representing the 50 states."

"Across the complete star there are 359,277,250 possible mathematical paths."

"Each path is one possible journey through the same larger structure."

"You can choose one path and make the finished star personal by adding your own six images."

Recommended classroom setup

Use one teacher computer connected to a projector or smartboard, or a large printed copy of the star. Students can do the mathematics on printed worksheets; individual devices are not required. If devices are available, the Star Creator can be used at the end so students can choose and personalize a path.

The question for this lesson

Do all starting positions create the same number of possible paths?

2. Lesson goals and teaching sequence

Learning goals

- Understand that reaching the same set of ending points does not mean having the same number of possible routes.
- Compare a near-edge starting position with a middle starting position and explain why the middle has more route-choice freedom inside one section.
- Understand why a physical corner of the complete star belongs to two adjoining sections and therefore receives paths from both.
- Use prediction, comparison, and explanation - not only arithmetic - to reason about a large path-counting system.

Suggested teaching sequence

1. Show the whole star: Point out the 50 starting points, 50 ending points, the five repeated sections, the modified Pascal's Triangle structure, and the 359,277,250 possible paths.

2. Predict: Ask which starting position students expect to create the most paths: near the edge, in the middle, or at a corner. Have students record a reason before seeing counts.

3. Compare: Show the near-edge and middle positions. Stress that both can reach all 10 ending positions in the section.

4. Explain: Guide students to route-choice freedom: a near-edge start must keep moving in one direction for many far-side endings, leaving fewer ways to vary the route.

5. Reveal the corner surprise: Return to the complete star and show that one physical corner is shared by two adjoining sections, so its one-section contribution is counted from both sections.

6. Create a personal connection: Connect the mathematical discovery back to the student experience: hundreds of millions of possibilities, one chosen route, and one personalized star.

Key vocabulary

Starting position: One of the positions from which a path begins in one section.

Ending position: One of the 10 positions where a path leaves that section.

Route-choice freedom: How many different step-by-step choices remain available while a path can still reach its ending position.

3. Near-edge vs. middle: the central comparison

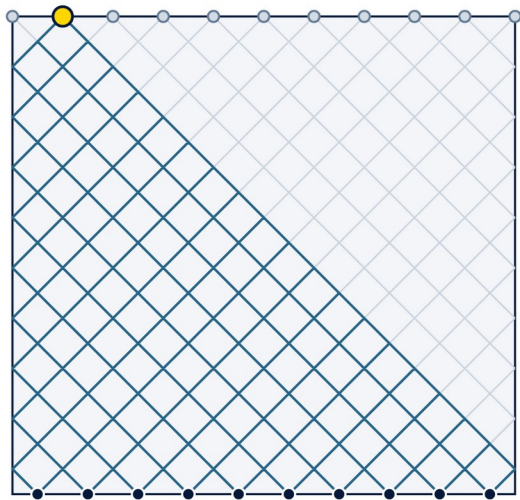
Both selected starts can reach all 10 ending positions. The question is not whether the endings are reachable. The question is how many different routes can reach them.

Simplified illustration — representative routes only

Selected start Valid branches from this start

Near-Edge Start

11 starting points · top



Same 10 ending points · bottom

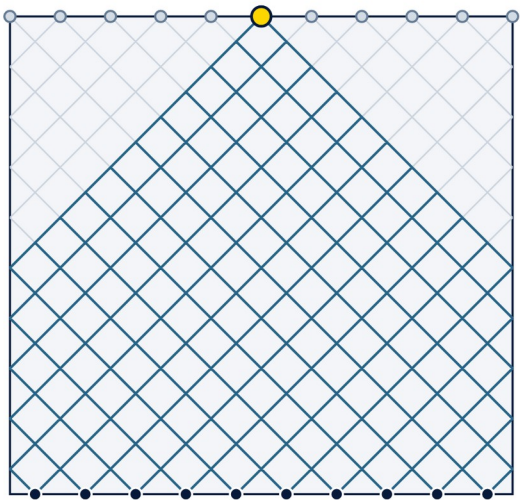
Can reach all 10 ending points, but reaching endings on the far side requires many steps to keep moving in that direction, leaving fewer ways to vary the route.

5,546,105

possible paths through one section

Middle Start

11 starting points · top



Same 10 ending points · bottom

Can reach all 10 ending points, with more freedom to vary left/right choices while still arriving at the same endings.

8,097,610

possible paths through one section

Same ending points. Different number of routes.

Both starting points can reach all 10 ending points.

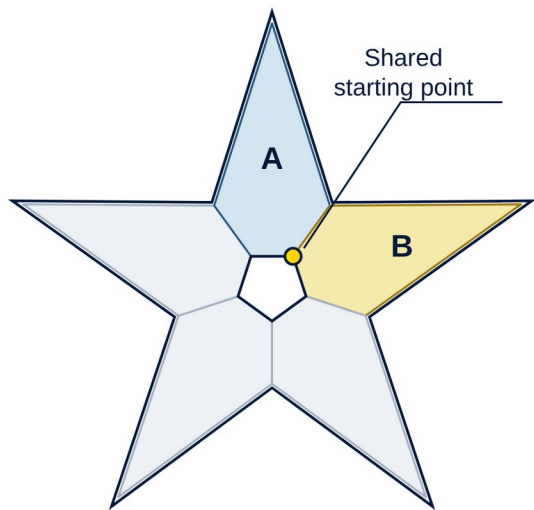
The middle can reach them through more different routes because it has more freedom to vary its left/right choices while still arriving at the same endings.

Teacher emphasis: the middle starting position has more freedom to vary left/right choices while still reaching the same endings. That additional freedom creates more distinct routes.

4. The full-star corner surprise

Inside one section, a corner has the fewest paths. In the complete star, however, that same physical corner belongs to two adjoining sections.

Simplified structural illustration — adjoining sections highlighted



Corner in complete star

4,194,280

+ 4,194,280

= 8,388,560

Middle starting position
8,097,610 paths

Difference: 290,950 more paths

Section A	Section B
4,194,280 paths	4,194,280 paths

One physical corner. Two adjoining sections.

The calculation

Corner, one section	4,194,280
Same physical corner, adjoining section	+ 4,194,280
Corner in complete star	= 8,388,560
Middle starting position	8,097,610
Result 8,388,560 - 8,097,610 = 290,950. The shared corner has 290,950 more paths than the middle in the complete star.	

5. Exact one-section counts and deeper teacher background

The exact one-section starting-position totals are symmetric. These numbers are teacher background; the web visual is usually enough for the first classroom explanation.

Starting position	One-section paths	Teaching note
1	4,194,280	corner
2	5,546,105	near-edge example
3	6,688,400	
4	7,496,735	
5	7,953,400	
6	8,097,610	middle
7	7,953,400	
8	7,496,735	
9	6,688,400	
10	5,546,105	near-edge example
11	4,194,280	corner

11 starts, 10 endings

1	2	3	4	5	6	7	8	9	10
5.546M	6.690M	7.506M	7.987M	8.199M	8.199M	7.987M	7.506M	6.690M	5.546M

Why the totals look different

The 11 starting-position totals and the 10 ending-position totals are two ways of grouping the same 71,855,450 paths in one section. Group by where a path starts and you get 11 totals; group the same paths by where they end and you get 10 totals. The grand total is identical.

6. Classroom prompts, short answers, and extensions

Q: If every starting point can reach every ending point, why can their total path counts still differ?

A: Because the number of distinct routes to those endings differs. Reachability is yes/no; route counting asks how many different sequences of choices work.

Q: Why does the middle have more possible routes inside one section?

A: It has more freedom to vary its left/right choices while still remaining able to reach the same endings.

Q: What changes when a corner is shared by two sections?

A: The same physical starting point receives path contributions from both adjoining sections.

Q: How is this connected to Pascal's Triangle?

A: The path structure uses a modified Pascal-style system in which choices branch and recombine. The full construction is more specialized, but the familiar Pascal idea gives students a useful mathematical reference point.

Optional extensions

- Prediction graph: rank the 11 starting positions before revealing the exact counts, then compare with the symmetric table.
- Percent comparison: calculate how much larger the middle is than the near-edge example inside one section.
- Symmetry check: identify which starting-position counts must match because the section is symmetric.
- Probability connection: discuss why a huge pool of possible paths makes exact path matches rare when students choose independently.
- Personal creation: after the mathematics, let students choose a path and personalize the six image spaces in their own star.

The larger educational idea

The Great American Star combines mathematics, art, America 250, and personal participation. The patriotic symbolism is not simply added to the artwork; it is embedded in the mathematics. A student moves from seeing one enormous system to choosing one individual path within it: "This one is mine."

359,277,250 possible paths. One American star. This one is mine.